
This item was submitted to [Loughborough's Research Repository](#) by the author.
Items in Figshare are protected by copyright, with all rights reserved, unless otherwise indicated.

Utilizing machine learning to understand and predict methane emissions in cattle farming with farm-scale environmental and biological variables

PLEASE CITE THE PUBLISHED VERSION

<https://doi.org/10.1109/ICCA62237.2024.10927962>

PUBLISHER

IEEE

VERSION

AM (Accepted Manuscript)

PUBLISHER STATEMENT

© 2024 IEEE. Personal use of this material is permitted. Permission from IEEE must be obtained for all other uses, in any current or future media, including reprinting/republishing this material for advertising or promotional purposes, creating new collective works, for resale or redistribution to servers or lists, or reuse of any copyrighted component of this work in other works.

LICENCE

All Rights Reserved

REPOSITORY RECORD

Partridge, Tom, Baihua Li, Bashar Alhnaity, and Qinggang Meng. 2024. "Utilizing Machine Learning to Understand and Predict Methane Emissions in Cattle Farming with Farm-scale Environmental and Biological Variables". Loughborough University. <https://hdl.handle.net/2134/28082495.v1>.

Utilizing Machine Learning to Understand and Predict Methane Emissions in Cattle Farming with Farm-Scale Environmental and Biological Variables

Tom Partridge
Department of Computer Science
Loughborough University
United Kingdom
T.Partridge@lboro.ac.uk

Qinggang Meng
Department of Computer Science
Loughborough University
United Kingdom
Q.Meng@lboro.ac.uk

Baihua Li
Department of Computer Science
Loughborough University
United Kingdom
B.Li@lboro.ac.uk

Bashar Alhnaity
Department of Computer Science
Loughborough University
United Kingdom
B.Alhnaity@lboro.ac.uk

Abstract— Cattle livestock contribute to climate change through enteric methane production, making it essential to identify and validate methods for reducing methane emissions. This research correlates GreenFeed cattle methane measurements with farm environment data using the North Wyke Farm Platform (NWFP), a heavily instrumented research facility in the UK. The disparate datasets are combined into a machine-learning-ready dataset capable of mapping methane emissions in grams per day and grams per kilogram of live weight gain. Predictive models are then developed and evaluated for methane prediction. Experimental results indicate that Gradient Boosting achieved the highest accuracy (g/day: $r=0.619$, RMSE=51.8; g/kg live weight gain: $r=0.562$, RMSE=65.9). Explainable AI methods are applied to quantify how a broad selection of farm and animal characteristics contribute to methanogenesis. This research provides valuable insights into methane reduction through machine learning and quantitative analytical methods.

Keywords—Methane emissions, Machine Learning, Livestock, Explainable AI

I. INTRODUCTION

When breaking down contributions of activities to greenhouse gas emissions, global livestock production contributes 14.5% of total anthropogenic emissions [4]. Of this, enteric fermentation then contributes 46% of greenhouse gas emissions [3]. Cattle are by far the greatest contributor to enteric fermentation due to their emissions per animal and their overall quantity. Enteric fermentation occurs inside ruminant livestock, processing its foodstuff into more readily digestible matter and releasing gasses as a byproduct, most notably methane which has a global warming potential 27 times greater than CO₂ per kilogram [8]. Cattle release their methane emissions from the mouth through regular eructation approximately every 90 seconds [10].

Ruminant cattle are major contributors to methane emissions, presenting a critical opportunity for mitigating climate change impacts [3]. The Intergovernmental Panel for Climate Change's (IPCC) methods for estimating methane emissions from farms are widely adopted but do not fully reflect current knowledge, practices, or the specific circumstances of various users. Dietary management and supplements have emerged as promising strategies to reduce methane emissions [6], though challenges remain in

identifying, verifying, and economically implementing effective solutions.

Methane management on farms involves two key components: emission estimation and reduction. While the IPCC-based models used for estimation are simple and accessible, they often lack accuracy and precision. In contrast, reducing emissions requires more sophisticated models incorporating detailed environmental variables to accurately simulate farm conditions. By refining these models, researchers can identify environmental factors that lower methane production. Once validated, these strategies can be adapted and applied across diverse farming operations.

This research intersects agricultural systems management, husbandry, and climate science, recognizing farms as complex systems where changes can have wide-ranging impacts. Farmers control various factors, such as cattle breeds and grass types, while also influencing aspects like soil pH through liming. Traditionally focused on productivity, farmers must now optimize these factors to reduce methane emissions, making it crucial to research which farm practices most effectively lower methane without compromising productivity.

This research employs Machine Learning (ML) to predict methane (CH₄) emissions based on farm conditions. The performance of the ML models is evaluated not only in terms of accuracy but also in relation to the explainability of input impact on the predicted methane results. Initially, a machine-learning-ready dataset is constructed by linking methane measurements for individual cattle to the corresponding farm conditions at the time and location of measurement. Multiple ML models are then trained and evaluated using this dataset to enable stakeholders to predict methane emissions for any given animal. Explainability methods are applied to analyze how the model utilizes its inputs, allowing stakeholders to explore potential interventions aimed at reducing methane production.

II. RELATED WORK

Machine learning has become an integral tool across various scientific disciplines due to its ability to process large datasets and uncover complex patterns. In climate science and agriculture, ML algorithms are particularly suited for analysing intricate systems, such as predicting weather patterns, optimizing crop yields, and assessing climate

impacts. By leveraging vast data sources, these techniques offer enhanced predictive capabilities and decision-making support. This has made ML indispensable for addressing global challenges like climate change and food security [14].

Methane emission estimation in cattle has developed increasingly accurate models. While generic estimates are available, models now incorporate specific inputs to improve accuracy. The IPCC provides formulas based on basic farm descriptors for accessible estimates, while more precise models, such as those formulated and collated by Niu et al. [12], focus on feed intake are accurate but impractical for farmers to measure in their standard processes. These models achieve performance of 16.6% measured as Root Mean Squared Error against sample set mean.

Dewhurst and Miller [1] briefly reviewed a range of studies investigating methane production with breed of cattle, finding little variation between breeds. Research is highly active for dietary control and supplementation opportunities. Most notably the 3-NOP inhibitor additive has found that when supplemented at 70.5mg/kg dry matter resulted in a methane 32.6% intensity reduction [2]. Hristov et al. [7] reviewed 3-NOP and other similar methods including electron sinks, oil & fat supplements, and tanniferous forages. They mentioned downsides that implementations can be cost prohibitive and can have diminishing efficacy over time.

Research on identifying and validating mitigation strategies requires appropriate datasets. Niu et al. [12] presented a meta-dataset compiled from multiple sources, providing diverse land and management types crucial for ensuring findings are generalizable. Traditionally, large-scale data collection has been hindered by methods like respiration chambers, which while accurate, disrupt natural emissions and are impractical for large-scale studies. GreenFeed sniffer machines offer a more efficient alternative, measuring methane and carbon dioxide from exhaled breath multiple times daily through an automated feed station. Validation studies have indicated that GreenFeed's accuracy is comparable to respiration chambers and mass flow controllers for both controlled-release samples [10] and cattle [5]. Hammond et Al. [5] cautioned that a significant number of measurements are required for reliable trend analysis, making small, short-term studies less likely to yield conclusive results.

The key opportunities in this research area relate to developing high quality estimates and interventions, and due to weaknesses in older data collection methods there are opportunities for improvement in all areas. Tailored estimates require development which accurately consider regional and strategic factors, and which are accessible to stakeholders. This research considers the discovery and verification of interventions to reduce methane. This is critical, particularly including consideration of uncertainty and compromise, such that trustworthy high-quality information may be promulgated through stakeholders.

III. THE NORTH WYKE FARM PLATFORM (NWFP)

The NWFP [11] is a research-focused cattle and sheep farm located in the south of the UK, renowned for its data collection and instrumentation. Established to support research initiatives, the NWFP offers a comprehensive array of data types collected regularly from 2008 to the present

(2024), with capabilities beyond that which could be expected from purely commercial sites, which gives this dataset rich analysis opportunities. Since 2017, NWFP has utilized automated GreedFeed machines to gather methane data from cattle to further enable climate research.

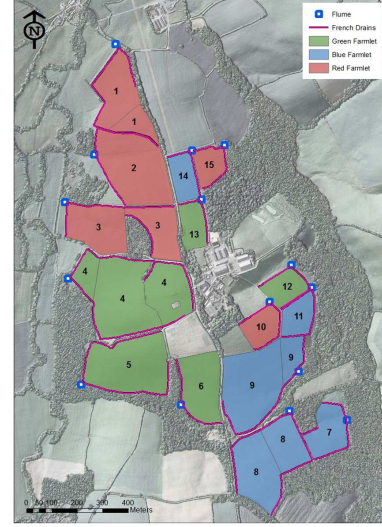


Figure 1 The North Wyke Farm Platform map. Separated into four farmlets, each with different crop strategies. Farmlets are Blue, Green, Red/Brown. A range of data is regularly collected from each field.

At the NWFP, each cattle is assigned to one of four treatments (farmlets) at the farm. Cattle are housed in winter months, and have continuous access to silage gathered from the farmlet they were on during summer. While housed, their methane and carbon dioxide emissions are autonomously measured several times per day using GreenFeed sniffing machines.

- Blue farmlet has a high-sugar / white clover mix.
- Green farmlet has permanent pasture.
- Red farmlet has high-sugar grasses.
- After April 2018, cattle which would be Red farmlet cattle are instead Brown farmlet. They are permanently housed and more intensively finished.

IV. DATASET GENERATION

To analyse farm factors in relation to methane emissions, data must be aligned with methane measurements. The NWFP records the farm's state, cattle IDs, and measurement times, allowing reconstruction of farm conditions for each methane reading. For categorical variables like cattle breed, alignment is straightforward. However, for continuous variables like weight, interpolation is required to estimate the animal's weight at the time of methane measurement using previous and subsequent weight records.

The resultant dataset is made of 260,901 sets of continuous and categorical variables over a range of different time periods whereby each methane measurement is related to a farm condition at the time and place of the methane measurement.

GreenFeed Data

Methane and carbon dioxide are measured against a unique cattle ID, recording time and duration of measurement. GreenFeed measurements recorded during the

winter after weaning until the animals are returned to pasture.

was equally represented in the dataset. For animal sex, 64% of animals were male, and 36% female.

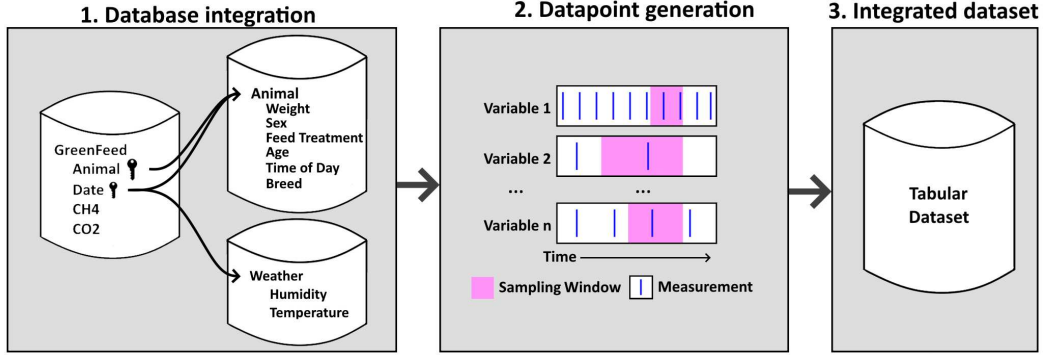


Figure 2 Integration of farm surveys and methane data using animal IDs and locations, followed by sampling with window size based on measurement frequency, to create a comprehensive livestock emissions dataset.

Weight

After weaning, cattle are regularly weighed. To align the cattle weight against GreenFeed measurements, a function was fit using the weights recorded at varying stages of the cattle's life for each cattle. This method accurately fits predicted weights against the measured weights with $r^2=0.974$.

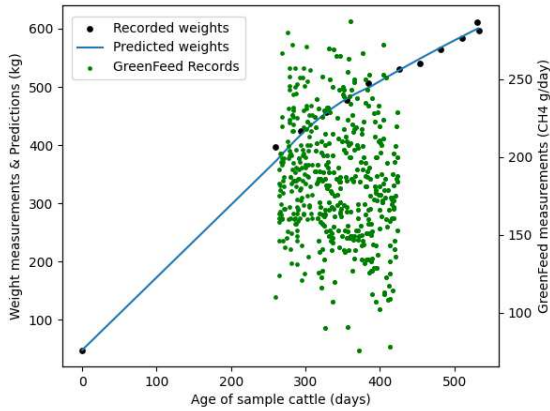


Figure 3 Weight records for a sample cattle, with the inferred weight fitment function and GreenFeed measurements.

Methane Intensity

Methane per kilogram of liveweight gain; this is extracted from the GreenFeed methane measurement and the weight estimation.

Age

Calculated as the difference between the date of methane sampling and the recorded date of birth of the animal.

Weather

All field sections have automated data collection every 15 minutes. All data matching the day of the GreenFeed record is mean averaged.

Categorical Variables

Farmlet, breed, sex, and location and other categorical information were encoded in one-hot manner. Location is known to the field or shed which the cattle was in. Breed code shorthand is used, where ST is Stabiliser, and STX is Stabiliser Cross, and CHX is Charolais Cross. Each breed

PRELIMINARY DATA ANALYSIS

The GreenFeed data itself is highly noisy, with measurement changing significantly between similar measurements. Some estimates of noise can be generated by

using the assumption that for any given cattle, there should be no change within the week. Therefore by taking the ratio of weekly standard deviations against overall standard deviation, the random variation can be estimated.

- For methane rate, the standard deviation of the dataset as a whole is 65.9, with mean average of 206.7. The average of the weekly standard deviations is 33.8. Therefore 51% of all deviation is unpredictable intra-week.
- For methane intensity, the standard deviation of the dataset is 116.2, with mean average of 254.7. The average of the weekly standard deviations is 44.1. Therefore 38% of all deviation is unpredictable intra-week.

V. DATASET CLEANING & DOMAIN-KNOWLEDGE BASED ATTRIBUTE SELECTION

There are a small number of outliers in the data which are often a magnitude of difference away from their neighbours. These are removed automatically by only including the lines in the dataset within 2.5 standard deviations (98.8% included).

Feature selection in machine learning models usually finds the most impactful input features to balance against the training strain incurred by increasing the quantity features input to a model. This then generates the model with the highest predictive performance. In this research, the explanation for how the model uses particular inputs is also desirable, such that they may be input to the model irrespective of their predictive impact. As the NWFP datasets contain a wide variety of features, domain knowledge was used to discern and exclude features which could not plausibly be causally related to the prediction. These features are presented in Figure 5 and Figure 6.

VI. MACHINE LEARNING PREDICTIVE MODEL

There are a range of successful model architectures which can each provide optimal performance. It is customary to fit and tune each of these models to determine which performs

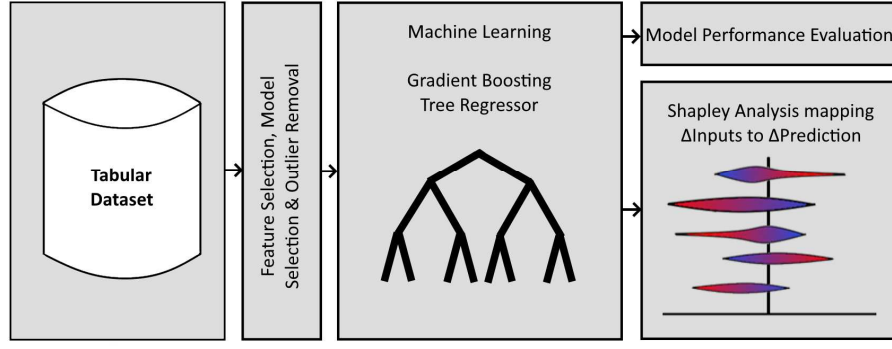


Figure 4 Process to transform the machine-learning ready dataset into an evaluated model with insight provided by Shapley Analysis.

optimally in the particular use case. Gradient Boosting regressor found consistently strong results [Table 1], so it was selected. This method builds an ensemble of weak decision trees, by sequentially minimizing the residual errors of prior models. Each new model is trained to correct the prediction errors of the existing ensemble, and the final prediction is a weighted sum of all models.

Machine learning models can memorize their training data, so part of the data was segregated to test performance on so that indicated performance is indicative. This was repeated using five-fold cross-validation so that the entire dataset was trained and tested upon while maintaining this segregation. Performance metrics were generated using the mean of these folds.

After manual hyperparameter tuning, the final gradient boosting regressor model used the following non-default hyperparameters. $n_estimators=200$, $learning_rate=0.3$. The model architecture was provided by the Sci-kit Learn library [13].

VII. RESULTS

The models, constrained by expected performance from preliminary analysis, predict methane generation of both formats with good performance. Rate prediction exceeds expectations, while intensity prediction falls slightly short, likely due to challenges in accurately capturing daily weight changes. Additionally, key inputs like feed intake, known to strongly correlate with methane production, were unavailable as features.

Model	Rate		Intensity	
	r	RMSE	r	RMSE
AdaBoost	0.612	52.6	0.561	66.0
Elastic Net	0.580	52.3	0.488	69.3
Deep Neural Network	0.613	51.7	0.549	66.6
Random Forest	0.616	52.0	0.561	65.7
Gradient Boost	0.619	51.8	0.562	65.9

Table 1 Hyperparameter-tuned performance results for a variety of model architectures for both prediction tasks. Correlation score (r) and prediction error (RMSE) are displayed.

MODEL EXPLAINABILITY

For analysis of groups of inputs, Shapley analysis can be used [9]. Shapley analysis is a method derived from cooperative game theory used to explain the contributions of individual features to the predictions made by machine

learning models. It calculates the Shapley value for each feature, which represents the average marginal contribution of that feature to the model's prediction.

For the plots, variables are sorted in overall contribution to model prediction from top to bottom. Amplitude of each plot is the quantity of samples with that impact. Red color

denotes that the input was high in the range while blue denotes that it was low. Categorical variables are one-hot encoded, where red means true, and blue means false.

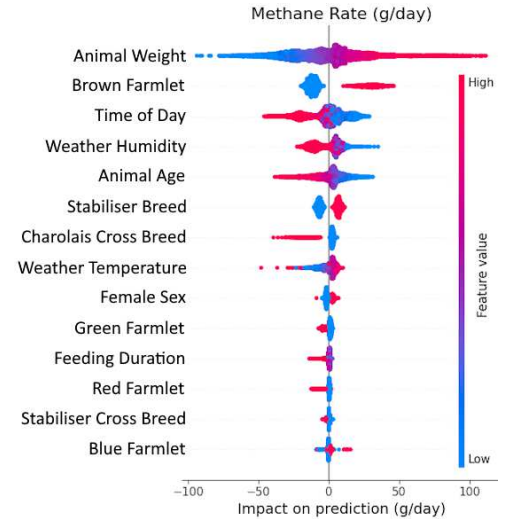


Figure 5 Shapley analysis of the methane rate prediction model.

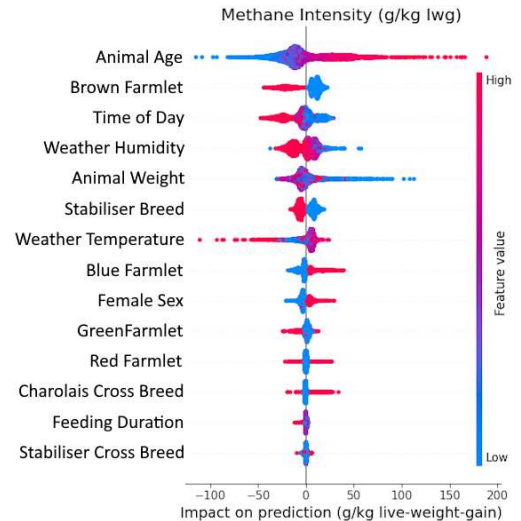


Figure 6 Shapley analysis of the methane intensity prediction model.

The methane rate and methane intensity Shapley analysis ascribes two different pictures to how methane should be considered depending on the use case of the cattle being raised. Most importantly the feed treatment strategies show that intensive feeding (Brown) will increase the rate at which methane is generated, due to increased metabolism, but that it may also encourage efficient feed usage in young cattle for mass growth.

The weight and age of the animals share information in the model, such that in both models, the less dominant feature is used as a tuning parameter against the more dominant feature. For methane rate, the weight of the animal dominates as this drives level of metabolism required to grow and maintain. However for methane intensity, the age of the animal strongly drives growth rate such that weight becomes less important. As the NWFP is a meat farm, it is unclear how these trends change once the animals have reached goal body weight, as might be the case for longer term milk farming.

Time of day and humidity are both similarly related against both rate and intensity, with relatively high model importance. Stabiliser breed finds a small amount of importance to the model, with increasing rate but decreasing intensity as found with intensive finishing. Of the remaining breeds, none were significantly impactful onto the result. Similarly, the sex of the animal was mostly unimportant, with females being slightly more methanogenic in both cases.

VIII. CONCLUSION

Following the need to further identify potential methods to reduce methane in cattle, we developed a process flow to correlate disparate datasets to allow analysis of relationships between variables and methane. For methane intensity, the age of the animal is the dominant factor, and intensive feeding is found to improve methane intensity. For methane rate, the weight of the animal was found to dominate the methane produced per day.

The analysis completed gives examples validity to larger-scale uses of GreenFeed to identify complex trends beyond estimation of methane from any given animal. Despite the advancements and opportunities which the GreenFeed machine provides, measurement precision remains a limiting factor to research.

With the development of high performing ML models, methods to analyse them need to be proven out and understood. In this case, Shapley has proven highly effective in explaining complex models, but shows the difficulty of disentangling inputs which carry shared information. This forms a challenge of experimental design whereby isolation of input variables becomes increasingly valuable.

ACKNOWLEDGMENTS

The project is funded by the UK Research Innovation and Engineering and Physical Sciences Research Council (EPSRC) grant (EP/Y00597X/1). We acknowledge data access of the North Wyke Farm Platform (BBS/E/RH/23NB0008, BBS/E/C/000I0320) at Rothamsted Research.

BIBLIOGRAPHY

- [1] Dewhurst, R. J., & Miller, G. (2019). Policy Brief: How do different livestock types, sizes and breeds differ in their greenhouse gas emissions? SRUC Dairy Research and Innovation Centre, SRUC Beef and Sheep Research Centre.
- [2] Dijkstra, E. K., Bannink, A., Pressman, E. M., Walker, N., Karagiannis, A., Gastelen, S. {., & Jan. (2023). A meta-analysis of effects of 3-nitrooxypropanol on methane production, yield, and intensity in dairy cattle. *Journal of Dairy Science*.
- [3] Food and Agriculture Organization. (2023). Pathways towards lower emissions – A global assessment of the greenhouse gas emissions and mitigation options from livestock agrifood systems. Food and Agriculture Organization.
- [4] Gerber, P., Steinfeld, H., Henderson, B., Mottet, A., Opio, C., Dijkman, J., . . . Tempio, G. (2013). *Tackling climate change through livestock – A global assessment of emissions and mitigation opportunities*. Food and Agriculture Organization of the United Nations (FAO).
- [5] Hammond, K., Humphries, D., Crompton, L., Green, C., & Reynolds, C. (2015). Methane emissions from cattle: Estimates from short-term measurements using a GreenFeed system compared with measurements obtained using respiration chambers or sulphur hexafluoride tracer. *Animal Feed Science and Technology*.
- [6] Hegarty, R. S., Passetti, R. A., Dittmer, K. M., Y. Wang, S., Emmet-Booth, J., Wollenberg, E., . . . Gurwick, N. (2021). An evaluation of emerging feed additives to reduce methane emissions from livestock. Edition 1. *A report coordinated by Climate Change, Agriculture and Food Security (CCAFS) and the New Zealand Agricultural Greenhouse Gas Research Centre (NZAGRC) initiative of the Global Research Alliance (GRA)*.
- [7] Hristov, A., Melgar, A., Wasson, D., & Arndt, C. (2022). Symposium review: Effective nutritional strategies to mitigate enteric methane in dairy cattle. *Journal of Dairy Science*.
- [8] Intergovernmental Panel on Climate Change (IPCC). (2023). *The Earth's Energy Budget, Climate Feedbacks and Climate Sensitivity*. Cambridge: Cambridge University Press.
- [9] Lundberg, S. M., & Lee, S.-I. (2017). *A unified approach to interpreting model predictions*. Curran Associates Inc.
- [10] McGinn, S. M., Coulombe, J.-F., & Beauchemin, K. A. (2021). Technical note: validation of the GreenFeed system for measuring enteric gas emissions from cattle. *Journal of Animal Science*.
- [11] Murray, P., Griffith, B., Orr, R., Shepherd, A., Blackwell, M., Hawkins, J., & Peukert, S. (2013). The North Wyke Farm Platform: A new UK national capability for research into sustainability of agricultural temperate grassland management. Rothamsted Research.
- [12] Niu, M., Kebreab, E., Hristov, A. N., Oh, J., Arndt, C., Bannink, A., . . . Hell. (2018). Prediction of enteric methane production, yield, and intensity in dairy cattle using an intercontinental database. *Global Change Biology*.
- [13] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., . . . others. (2011). *Scikit-learn: Machine learning in Python*. Journal of machine learning research.
- [14] Rolnick, D., Donti, P. L., Kaack, L. H., Kochanski, K., Lacoste, A., Sankaran, K., . . . Mukkav. (2023). Tackling Climate Change with Machine Learning. *Association for Computing Machinery*.